

Santosh Adabala

Machine Learning Engineer | Healthcare NLP | Applied AI
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Professional Summary

Machine Learning Engineer with 5+ years of experience building applied ML, NLP, and predictive analytics solutions for healthcare and insurance data. Focused on healthcare NLP, claims risk scoring, denial prediction, fraud-risk detection, anomaly detection, and ML-ready data pipelines. Hands-on with Python, SQL, Scikit-learn, PyTorch, TensorFlow, ClinicalBERT, PySpark, Azure, AWS, Databricks, MLflow, and model evaluation workflows.

Technical Skills

Languages: Python, SQL, R, Scala

ML/AI: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, ClinicalBERT, Transformers, Hugging Face, RAG, NLP, NER, Anomaly Detection

Evaluation/MLOps: Precision, Recall, F1, AUC, MLflow, Model Monitoring, Model Deployment, Batch Inference, CI/CD

Data/Cloud: PySpark, Databricks, Airflow, Azure Data Factory, Azure Synapse, Azure SQL, AWS SageMaker, AWS EMR, S3, SQL Server, Snowflake, BigQuery

Analytics/Tools: Power BI, Tableau, DAX, Git, GitHub Actions, Docker, Terraform, HIPAA, PHI/PII, Data Quality

Professional Experience

Machine Learning Engineer Aug 2024 – Present
Blue Cross Blue Shield of Colorado Denver, CO

- Built healthcare NLP workflows with ClinicalBERT, Python, and Scikit-learn to extract patterns from clinical notes, claims narratives, and prior authorization text, reducing manual clinical review effort by 25%.
- Developed Python and SQL-based models to prioritize high-risk claims, predict likely denials, and surface high-cost cases for utilization review teams, improving claims triage accuracy by 18% based on precision/recall evaluation.
- Built HIPAA-compliant ML data pipelines using Azure Data Factory, Azure Synapse, Azure SQL, SQL Server, and Python to process claims, eligibility, provider, and pre-certification data from 6+ payer systems, reducing manual data preparation by 30%.
- Created Power BI dashboards and model-monitoring reports to track denial rates, turnaround time, appeals success, utilization trends, provider performance, and data quality checks, reducing recurring reporting turnaround time by 40%.

Machine Learning Engineer Jul 2019 – Aug 2022
Accenture – Sun Life Insurance Client India

- Built predictive models with Random Forest, XGBoost, Logistic Regression, and Scikit-learn to analyze claims, policyholder behavior, risk patterns, and claim approval trends, improving risk segmentation accuracy by 20% using F1 and AUC-based evaluation.
- Developed anomaly detection logic on claims and transaction data to flag suspicious billing activity, unusual claim frequency, and high-risk policy behavior, reducing manual review effort by 25%.
- Prepared ML-ready datasets from policy, premium, claims, customer, and payment data using Azure Data Factory, Azure Synapse, Azure SQL, PySpark, and Python, improving ingestion throughput by 50% and reducing reporting costs by 31%.
- Deployed Azure Databricks pipelines with data quality checks, logging, alerting, and SLA monitoring for recurring insurance analytics and model input datasets, improving SLA adherence by 22%.

Machine Learning Projects

AlignLLM: LLM Alignment Pipeline on AWS | SageMaker, LoRA/QLoRA, DPO, RLHF, AWS Glue, Terraform

- Built an LLM alignment pipeline for a 7B chat model using SFT, DPO, RLHF, and LoRA/QLoRA; trained on AWS SageMaker, processed datasets with Glue and Athena, tracked experiments in W&B, and compared model behavior using MT-Bench, MMLU, TruthfulQA, and lm-evaluation-harness.

LLM Model Compression and Inference Optimization | PyTorch, Transformers, ONNX

- Fine-tuned and compressed transformer models using knowledge distillation and ONNX export, reducing model size from 110M to 65M parameters and improving inference latency from 39ms to 11ms.

Distributed ML Pipeline for Document Intelligence | AWS EMR, S3, PySpark, MLflow

- Built a PySpark-based document processing pipeline on AWS EMR and S3, improving throughput to 4,350 docs/sec by tuning Spark transformations, partitioning strategy, and batch processing logic.

Education

M.S. Data Science, University of Colorado Boulder

B.Tech. Electronics and Communication Engineering, JNTU Kakinada

Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Mining, Statistics, Big Data Analytics

Certifications

AWS ML Engineer – Associate | AWS Solutions Architect – Associate | NVIDIA AI Anomaly Detection | Azure DP-900